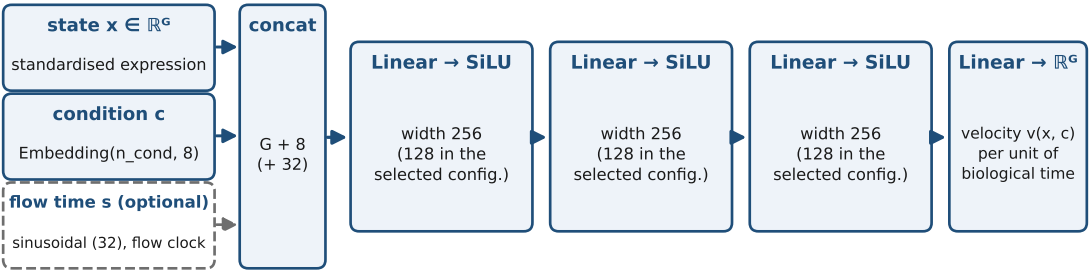
Velocity-field architecture (selected model: autonomous field, spline targets)

parameters: 44,464 at width 128 (selected GRN configuration) · 154,416 at width 256



Training (spline targets)

sample (x(u), ẋ(u), c) from the OT-chained splines
loss $\|v(x(u) + \sigma_{min}\,\varepsilon, c) - \dot{x}(u)\|^2$
Adam 10^{-3} · gradient clip 1.0 · batch 256
EMA 0.999 weights; early stop on the validation energy distance; no refit

Inference (biological horizon)

$x(t_b) = x(t_a) + \int v(x, c)\,dt$ over $\Delta t = t_b - t_a$
RK4, 20 steps per unit of biological time
any horizon: interpolation and forecast use the same field; controls are scored in the same standardised gene space